

ANALYTICS CONSULTING REPORT



Shield Insurance

Insurance Analytics Dashboard

From Business Problem to Live Dashboard — A Complete Analytics Engagement Report

Client	Shield Insurance (Pilot Project)
Project Type	Business Intelligence & Dashboard Development
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Program	Codebasics Virtual Internship
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Classification	Confidential — Client Use Only

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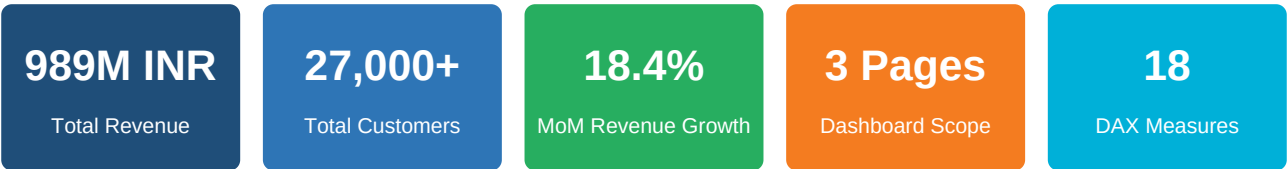
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Executive Summary

Shield Insurance, a growing insurance provider, engaged our analytics team to design and deliver a comprehensive business intelligence solution addressing critical gaps in revenue visibility, customer analytics, and sales channel performance monitoring. This report documents the complete engagement — from initial client requirements through solution delivery and business impact assessment.

The delivered solution is a three-page interactive Power BI dashboard covering the General View, Sales Mode Analysis, and Age Group Analysis — with 17 features, 18 custom DAX measures, and a structured data model built on 5 source tables. The dashboard enables leadership to make faster, evidence-based decisions without manual data extraction.



Key Finding: The 31-40 age group represents the highest-value customer segment across all metrics. Offline-Agent channels dominate revenue (55.4%) but digital channels are accelerating — signalling a strategic inflection point for Shield Insurance.

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Client Background & Business Problem

2.1 About Shield Insurance

Shield Insurance is a mid-sized insurance provider operating across multiple Indian cities through four distinct sales channels: Offline-Agent, Offline-Direct, Online-App, and Online-Website. The business serves customers across all age demographics and offers a portfolio of insurance policies at varying premium levels.

2.2 The Business Challenge

Despite operating at scale, Shield Insurance lacked a unified analytics infrastructure. Business leadership faced three persistent challenges that impaired decision-making speed and accuracy:

Revenue Visibility Gap	Monthly revenue reporting required hours of manual Excel aggregation. Leadership had no real-time visibility into daily revenue trends or month-over-month growth rates, making it impossible to detect performance issues early.
Customer Demographics Unknown	The business had no systematic view of which age groups or geographies were driving customer acquisition and revenue. Marketing and product decisions were made without demographic grounding.
Sales Channel Silos	Four sales channels operated independently with no unified performance comparison. Leadership could not determine which channels were growing, declining, or most cost-effective — preventing informed channel investment decisions.

2.3 Project Objective

The objective of this pilot engagement was to deliver a production-ready Power BI dashboard that provides Shield Insurance's leadership with on-demand, self-service analytics — eliminating manual reporting and enabling faster, data-driven decisions across revenue, customer demographics, and sales channel performance.

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Project Scope & Requirements

3.1 Requirements Extraction Process

Client requirements were delivered via a formal project brief email. Our team extracted and structured 17 discrete features from the narrative requirements, classifying each by priority level (High / Low) and dashboard page. This feature list served as the contractual scope baseline for the engagement.

3.2 Feature List — General View

#	Feature	Priority	Page
1	Total Customers, Total Revenue, Daily Revenue Growth, Daily Customer Growth as KPI cards	High	General View
2	Month-over-Month change % on all key metrics	High	General View
3	Customer segmentation by age groups: 18-24, 25-30, 31-40, 41-50, 51-65, 65+	High	General View
4	Total Revenue split by Age Group and City	High	General View
5	Total Customers split by Age Group and City	High	General View
6	Daily Customers growth trend by month	Low	General View
7	Daily Revenue growth trend by month	Low	General View
8	Toggle switch between Revenue and Customer trend graphs	Low	General View
9	Filters on Sales Mode, Age Group, City, Month, Policy ID	High	All Pages
10	Separate page for Sales Mode Analysis	High	Sales Mode
11	Total Customers split percentage by Sales Mode	High	Sales Mode
12	Total Revenue split percentage by Sales Mode	High	Sales Mode
13	Trend of Sales Mode performance over month	Low	Sales Mode
14	Separate page for Age Group Analysis	High	Age Group
15	Age Group vs Expected Settlement percentage	High	Age Group
16	Age Group vs Sales Mode preference	Low	Age Group
17	Age Group vs Policy Preference	High	Age Group

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Stakeholder Feedback & Incorporated Changes

4.1 Mockup Review Process

Following requirements confirmation, our team developed a three-page wireframe mockup representing the proposed dashboard layout, visual types, and information hierarchy. The mockup was submitted to the client for review prior to any development work — ensuring alignment before engineering investment.

4.2 Client Response

"I appreciate the mock-up that you created for us. It was clear that you put a lot of effort and creativity into the design, and we were all very impressed with the end result... I did make a few changes to the mock-up to better align with our specific needs." — Mathew (Client Stakeholder, Shield Insurance)

4.3 Key Design Changes Incorporated

Layout Restructuring	The client revised the General View layout, placing Revenue Split and Customer Split as side-by-side tables rather than stacked verticals. The trend chart was repositioned to the right panel alongside the Filters section.
Customer Segmentation Addition	A combined City × Age Group segmentation table was added to the General View page, providing a cross-dimensional view not present in the original mockup.
Age Group Page Redesign	The Age Group Analysis page was updated to include a trend chart by age group over months (multi-line) and a horizontal bar chart of Age Group vs No. of Customers — replacing the original single bar chart layout.
Sales Mode Visualization	The Sales Mode page was simplified to two large split percentage visuals (Customer and Revenue) plus the trend line — removing the KPI card row that was in the original mockup.
Branding Alignment	Shield Insurance logo, color palette (navy, cyan, orange) and page title style were standardized across all pages based on client brand guidelines.

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Solution Architecture

5.1 Data Sources

The solution is built on five CSV source files provided by the client, structured into a Star Schema data model in Power BI:

Table	Type	Key Columns	Business Purpose
dim_customer	Dimension	customer_code, dob, city	Customer demographics & age group derivation
dim_date	Dimension	date, mmm_yy, day_type, week_no	Time intelligence & monthly trend analysis
dim_policies	Dimension	policy_id, base_cover, base_premium	Policy catalog & premium benchmarks
fact_premiums	Fact (Core)	date, customer_code, policy_id, sales_mode, final_premium	All policy transactions — primary revenue source
fact_settlements	Standalone	age, settlement%	Actuarial settlement risk by age

5.2 Data Quality Issue Resolved

Issue: Power BI auto-detected the *settlement%* column as a Percentage data type, multiplying all values by 100 (34.15% became 3415%). **Resolution:** Data type changed to Decimal Number in Power Query before loading — a standard data quality step for percentage columns stored as decimals in source files.

5.3 Data Model — Star Schema

fact_premiums serves as the central fact table connected to three dimension tables. fact_settlements is maintained as a standalone table, queried independently for the age group settlement analysis without requiring a direct relationship to the transaction data.

Relationship	Cardinality	Join Key
dim_customer → fact_premiums	One-to-Many (1:*)	customer_code
dim_date → fact_premiums	One-to-Many (1:*)	date
dim_policies → fact_premiums	One-to-Many (1:*)	policy_id
fact_settlements	Standalone — no relationship	Used via AVERAGE measure

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Dashboard Walkthrough

6.1 Home Page — Navigation Hub

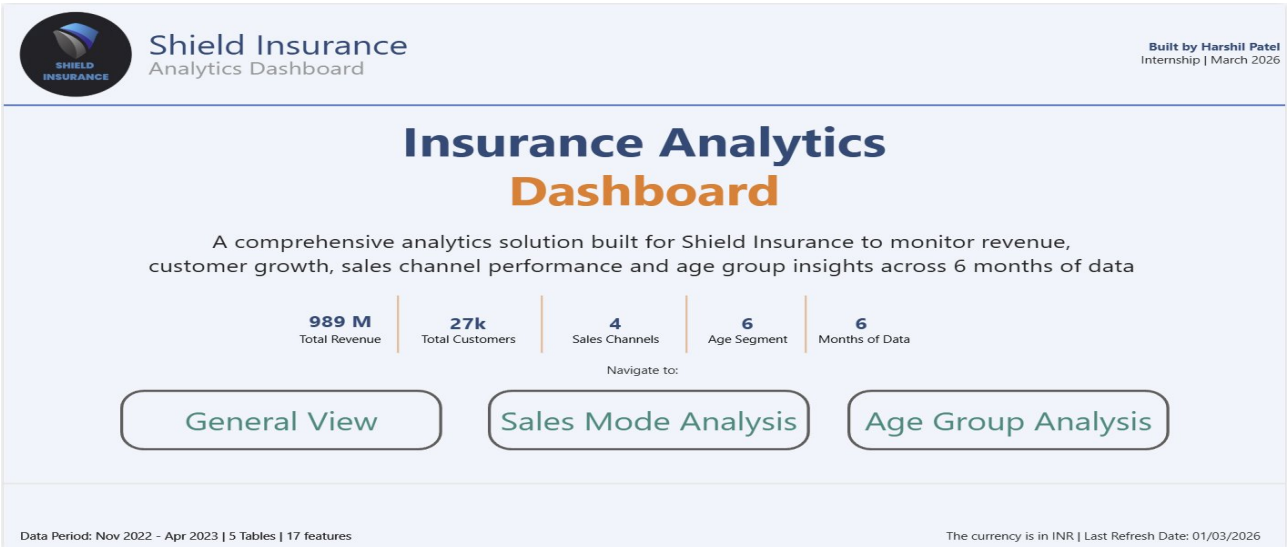


Figure 1 — Home Page: Provides immediate context (989M revenue, 27K customers, 6 months of data) and one-click navigation to each analytical page.

The home page was designed as the client's entry point into the dashboard. Three navigation buttons eliminate the need for stakeholders to navigate Power BI tabs manually — a critical UX decision for non-technical executives. Quick-stat callouts provide an at-a-glance snapshot without requiring page navigation.

6.2 General View — Business Pulse

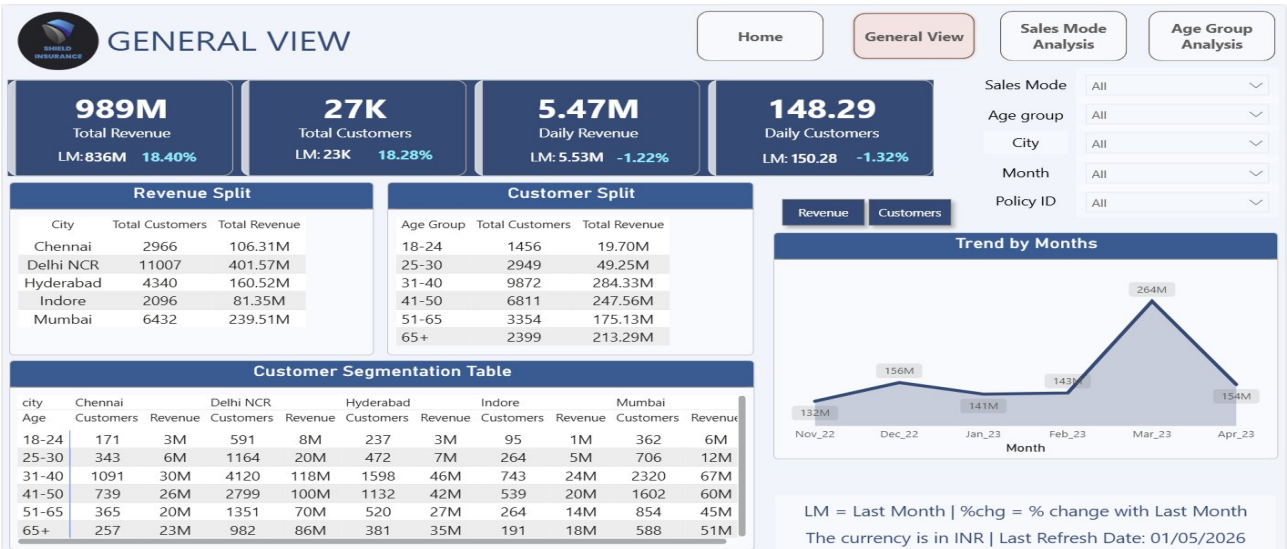


Figure 2 — General View: KPI cards with LM/MoM comparisons, split tables, trend chart with toggle, and customer segmentation matrix.

The General View consolidates the four most critical business metrics into KPI cards at the top: Total Revenue (989M), Total Customers (27K), Daily Revenue (5.47M), and Daily Customers (148). Each card surfaces the Last Month (LM) value and the month-over-month percentage change — enabling leadership to assess performance trajectory at a glance rather than comparing isolated period values.

6.3 Sales Mode Analysis — Channel Intelligence

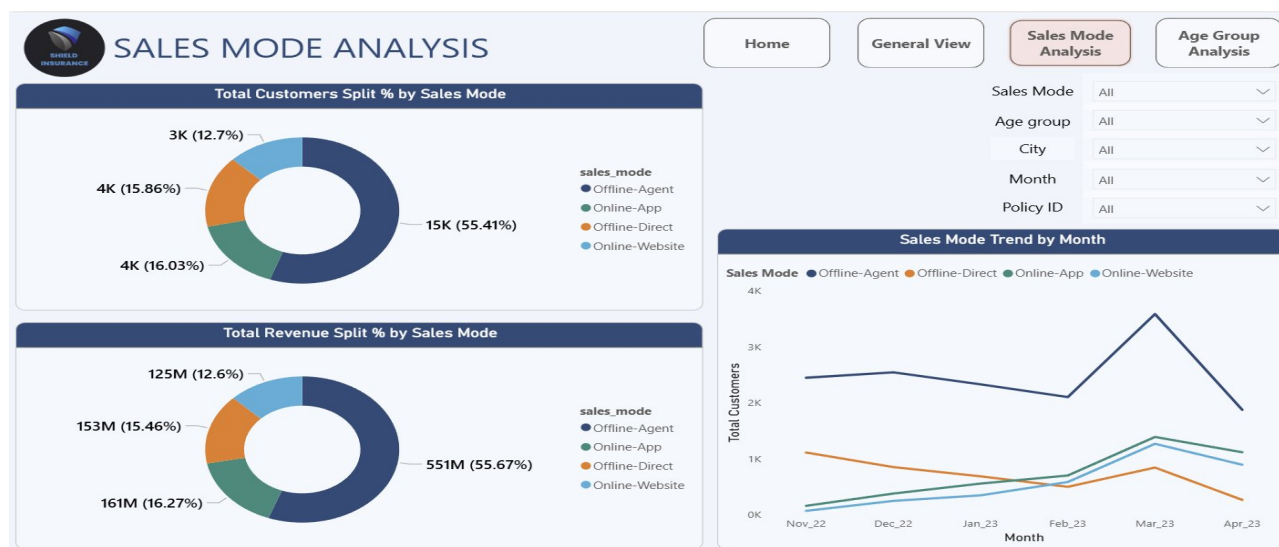


Figure 3 — Sales Mode Analysis: Customer and Revenue split by channel, and monthly trend across all four sales modes.

6.4 Age Group Analysis — Demographic Intelligence



Figure 4 — Age Group Analysis: Policy preference, expected settlement risk, sales mode preference, and customer trend — all segmented by age group.

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Key Findings & Business Insights

The following five findings represent the most actionable insights surfaced by the analytics dashboard. Each finding is grounded in the data and carries direct strategic implications for Shield Insurance's leadership.

1

Revenue Growth is Strong — But the Pace is Slowing

Total revenue grew +18.4% MoM, but Daily Revenue declined -1.22%. Growth was concentrated in March (264M) and did not sustain into April (154M). Leadership should investigate the March driver and build a repeatable strategy.

2

Delhi NCR is the Undisputed Revenue Capital

Delhi NCR generates 401.57M across 11,007 customers — nearly double Mumbai (239.51M). Agent deployment, marketing spend, and product development should be prioritized here first.

3

The 31-40 Age Group is the Core Revenue Engine

The 31-40 segment leads every dimension: highest count (9,872 customers), highest revenue (284.33M), broadest policy preference, and highest engagement across all sales channels. Retention programs should center on this segment.

4

Offline-Agent Dominates — But Digital is Accelerating

Offline-Agent holds 55.41% of customers and 55.67% of revenue. However, Online-App and Online-Website showed steeper growth in March. This signals an early digital transition requiring proactive infrastructure investment.

5

65+ Settlement Rate is an Actuarial Risk Signal

The Expected Settlement curve reaches 83% for the 65+ group — meaning 83 of every 100 policies result in a claim. Premium pricing for this age band warrants immediate actuarial review to ensure portfolio sustainability.

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Business Impact & Recommendations

8.1 Immediate Business Impact

The delivered dashboard eliminates the primary operational friction points identified at project inception. The following table summarises the before/after state for each business challenge:

Business Challenge	Before Dashboard	After Dashboard
Revenue Visibility	Manual Excel aggregation, hours per report, no daily visibility	Real-time KPI cards with daily granularity and MoM % change — instant access
Customer Demographics	No systematic demographic view — decisions made on intuition	City × Age Group segmentation matrix with revenue and customer breakdowns
Sales Channel Performance	4 channels operating in silos — no unified comparison possible	Side-by-side split % charts and trend lines across all 4 channels
Executive Reporting	Static reports prepared manually — delivered days after period close	Self-service dashboard — leadership accesses live data independently

8.2 Strategic Recommendations

Immediate (0-30 days)

- Investigate the March revenue spike: identify the specific driver (campaign, policy promotion, agent activity) and build a repeatable strategy.
- Review premium pricing for the 65+ age group against the 83% settlement rate — engage the actuarial team immediately.
- Concentrate agent and marketing resources in Delhi NCR where ROI is highest.

Short-Term (30-90 days)

- Develop targeted retention programs for the 31-40 age segment — the core revenue engine.
- Begin investment in Online-App and Online-Website infrastructure to capture accelerating digital demand before channel share shifts significantly.
- Expand dashboard to include policy-level profitability analysis.

Medium-Term (90+ days)

- Build predictive churn models for the 51-65 age group (approaching settlement phase).
- Develop city-level expansion analysis — identify next priority cities beyond Delhi NCR.
- Automate monthly reporting via Power BI scheduled refresh and email subscriptions.

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Appendix — DAX Measures & Data Model

A.1 Complete DAX Measures Reference

Core Measures

Measure Name	DAX Formula
Total Revenue	SUM(fact_premiums[final_premium_amt(INR)])
Total Customers	DISTINCTCOUNT(fact_premiums[customer_code])
No of Days	DATEDIFF(MIN(dim_date[date]), MAX(dim_date[date]), DAY) + 1
Daily Revenue	DIVIDE([Total Revenue], [No of Days])
Daily Customers	DIVIDE([Total Customers], [No of Days])

Last Month (LM) Measures

Measure Name	DAX Formula
Revenue LM	CALCULATE([Total Revenue], DATEADD(dim_date[date], -1, MONTH))
Customers LM	CALCULATE([Total Customers], DATEADD(dim_date[date], -1, MONTH))
Daily Revenue LM	CALCULATE([Daily Revenue], DATEADD(dim_date[date], -1, MONTH))
Daily Customers LM	CALCULATE([Daily Customers], DATEADD(dim_date[date], -1, MONTH))

MoM % Change Measures

Measure Name	DAX Formula
Revenue % Change (MoM)	DIVIDE([Total Revenue] - [Revenue LM], [Revenue LM])
Customers % Change (MoM)	DIVIDE([Total Customers] - [Customers LM], [Customers LM])
Daily Revenue Growth %	DIVIDE([Daily Revenue] - [Daily Revenue LM], [Daily Revenue LM])
Daily Customer Growth %	DIVIDE([Daily Customers] - [Daily Customers LM], [Daily Customers LM])

Split & Settlement Measures

Measure Name	DAX Formula
Revenue Split %	DIVIDE([Total Revenue], CALCULATE([Total Revenue], ALL(fact_premiums)))
Customers Split %	DIVIDE([Total Customers], CALCULATE([Total Customers], ALL(fact_premiums)))

Measure Name	DAX Formula
Expected Settlement %	AVERAGE(fact_settlements[settlement %])

A.2 Technology Stack

Tool / Technology	Version	Role in Project
Power BI Desktop	Latest	Dashboard development, DAX, data modelling
DAX (Data Analysis Expressions)	—	18 custom measures + 2 calculated columns
Power Query (M Language)	—	Data loading, type correction, transformations
Python 3.11	3.11+	Mockup PDF, presentation, background image generation
ReportLab	4.x	PDF report and mockup generation
pptxgenjs	3.x	Programmatic PPTX presentation generation
MySQL	8.0	Star Schema design and data modelling concepts
Microsoft Excel	365	Feature list, DAX metrics planning

This report was prepared by Harshil Patel as part of the Codebasics Virtual Internship program (designed by Dhaval Patel and Hemanand Vadivel). Shield Insurance is a fictional client created for educational purposes. All data and figures are from the pilot dataset provided as part of the internship curriculum.